

Problem Set 4

*Each student must submit your own **HANDWRITTEN** work. Turn in a paper copy at the beginning of lecture on 10/12/2026. Please write your name on each page and make sure your problem set is stapled. Students are allowed to work in groups, but each student must turn in their own set of answers. No help from outside of the course is allowed. To receive full credit for a problem, all work must be shown. Partial credit will be awarded. All problems are weighted evenly.*

1. Suppose that the production of airframes is instead characterized by a production function where $Q(K, L) = KL$.
 - (a) Find the marginal products of capital and labor for this production function.
 - (b) Write the optimality condition that this firm's choice of labor and capital must satisfy in order to be cost-minimizing.
 - (c) Using the optimality condition and the production function, find the firm's conditional input demand functions for labor and capital.

- (d) Now suppose the the price of labor is \$10/unit and the price of capital is \$1/unit. Find the cost minimizing combination of labor and capital for an airframe manufacturer that wants to produce 121,000 airframes.

2. The processing of payroll for the 10,000 workers in a large firm can either be done using 1 hour of computer time (denoted by K) and no clerks or with 10 hours of clerical time (denoted by L) and no computer time. Computers and clerks are perfect substitutes; for example, the firm could also process its payroll using $1/2$ hour of computer time and 5 hours of clerical time.
- (a) With L on the x -axis and K on the y -axis, sketch the isoquant that shows all combinations of clerical time and computer time that allows the firm to process the payroll for 10,000 workers.

Now suppose computer time costs \$5 per hour and clerical time costs \$7.50 per hour.

- (b) On the graph from part (a), draw at least three isocost lines for the firm for various levels of total cost. Also, indicate the slope of the isocost lines.

- (c) What are the cost-minimizing choices of L and K ?
- (d) What is the minimized total cost of processing the payroll?
- (e) Explain, in terms of the optimality condition, why the firm chooses to use only one input, rather than some of each input.
- (f) Suppose that the price of clerical time remains at \$7.50 per hour. How high would the price of an hour of computer time have to be before the firm would find it worthwhile to use only clerks to process the payroll?

3. Consider a production function with the three inputs K capital, L labor, M materials which have the prices $r = 1$, $w = 1$, and $m = 1$ respectively. The firm's production function is given by $Q = K^{\frac{1}{3}}L^{\frac{1}{3}}M^{\frac{1}{3}}$. Solve for the firm's cost minimizing choice of capital, labor, and materials if the firm wants to produce 12 units of output.